

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Amazon Web Services, VA -- station KHEF, America/New_York
Committed pair OSM 1280340214 -> 1287260554
Amazon Web Services / Amazon Web Services
Facade gap 72.7 m between the two halls
Weather record 42,554 real hourly records from KHEF
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.3903 C at 0 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2023-04-11 (crossing)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 1 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 3 of 24 hours with 1 mode change. The reactive incumbent took 12 hours -- 9 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 11 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 3 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 12 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
the incumbent broke its own switch budget 1 time(s) to stay safe
11 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	5.175	18.0	5.007	1.043
01	mechanical	yes	switch budget	5.302	18.0	4.124	1.352
02	mechanical	yes	switch budget	4.472	18.0	2.454	1.249
03	mechanical	yes	switch budget	4.752	18.0	3.014	1.207
04	mechanical	yes	switch budget	3.362	18.0	1.344	1.153
05	mechanical	yes	switch budget	1.997	18.0	0.234	1.119
06	mechanical	yes	switch budget	2.497	18.0	-0.376	1.425
07	mechanical	yes	switch budget	5.027	18.0	4.124	1.731
08	mechanical	yes	switch budget	8.917	18.0	9.124	2.460
09	mechanical	yes	switch budget	11.556	18.0	12.781	2.748

10	mechanical	yes	switch budget	16.417	18.0	16.904	3.016
11	mechanical	no	dry-bulb	19.076	18.0	20.001	2.233
12	mechanical	no	dry-bulb	21.697	18.0	23.014	1.846
13	mechanical	no	dry-bulb	23.474	18.0	24.235	1.512
14	mechanical	no	dry-bulb	24.894	18.0	24.821	1.398
15	mechanical	no	dry-bulb	25.799	18.0	25.739	1.198
16	mechanical	no	dry-bulb	25.693	18.0	25.016	1.362
17	mechanical	no	dry-bulb	24.981	18.0	24.446	1.431
18	mechanical	no	dry-bulb	24.476	18.0	22.783	1.753
19	mechanical	no	dry-bulb	21.967	18.0	18.564	1.962
20	mechanical	no	dry-bulb	20.307	18.0	15.794	2.075
21	FREE	yes	-	17.807	18.0	12.454	1.924
22	FREE	yes	-	15.302	18.0	11.344	1.805
23	FREE	yes	-	12.527	18.0	8.564	1.458

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.175 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.302 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.472 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.752 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.362 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.997 C

against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.497 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.027 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.917 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.556 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.417 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.076 C against a 18.0 C limit, so it fails by 1.076 C. It would take a limit 1.076 C higher, or a bound 1.076 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.697 C against a 18.0 C limit, so it fails by 3.697 C. It would take a limit 3.697 C higher, or a bound 3.697 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.474 C against a 18.0 C limit, so it fails by 5.474 C. It would take a limit 5.474 C higher, or a bound 5.474 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.894 C against a 18.0 C limit, so it fails by 6.894 C. It would take a limit 6.894 C higher, or a bound 6.894 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.799 C against a 18.0 C limit, so it fails by 7.799 C. It would take a limit 7.799 C higher, or a bound 7.799 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.693 C against a 18.0 C limit, so it fails by 7.693 C. It would take a limit 7.693 C higher, or a bound 7.693 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.981 C against a 18.0 C limit, so it fails by 6.981 C. It would take a limit 6.981 C higher, or a bound 6.981 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.476 C against a 18.0 C limit, so it fails by 6.476 C. It would take a limit 6.476 C higher, or a bound 6.476 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.967 C against a 18.0 C limit, so it fails by 3.967 C. It would take a limit 3.967 C higher, or a bound 3.967 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.307 C against a 18.0 C limit, so it fails by 2.307 C. It would take a limit 2.307 C higher, or a bound 2.307 C tighter, to change this hour.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.807 C, which is 0.193 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.924 C, and the margin is measured, not chosen: 1.575 C of group-conditional forecast error for this hour of day, plus 0.3488 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.302 C, which is 2.698 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.805 C, and the margin is measured, not chosen: 1.456 C of group-conditional forecast error for this hour of day, plus 0.3488 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.527 C, which is 5.473 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.458 C, and the margin is measured, not chosen: 1.109 C of group-conditional forecast error for this hour of day, plus 0.3488 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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